# Project 2: Turbojet Engine Design/Off-Design Performance Analysis

Riley J. Kyle*
*Missouri University of Science and Technology, Rolla, Missouri, 65409, United States*

Dr. David W. Riggins†
*Missouri University of Science and Technology, Rolla, Missouri, 65409, United States*

**The purpose of this project is to plot the performance three space of a fixed-area missile-class engine with a simple converging nozzle. This will be done by varying the fuel flow rate from 20% to 500% the on design fuel rate, varying the Mach number from 0 to 2.5. Additionally, this will be done at sea level, 4000 m, 8000 m, and 12000 m altitude.**

## Nomenclature

$T$ = temperature $[K]$

$P$ = pressure $[Pa]$

$\rho$ = density $\left[kg/m^3\right]$

$T_t$ = total temperature $[K]$

$P_t$ = total pressure $[Pa]$

$\tau$ = $\frac{T_{t,out}}{T_{t,in}}$

$\pi$ = $\frac{P_{t,out}}{P_{t,in}}$

$\eta$ = isentropic efficiency

$u$ = velocity $[m/s]$

$M$ = mach number

$A$ = area $\left[m^2\right]$

Subscripts

0 = freestream

$d$ = diffuser

$c$ = compressor

$b$ = burner

$t$ = turbine

*Aerospace Engineering Student, MAE
†Professor of Mechanical and Aerospace Engineering, MAE

$n$ = nozzle

# I. Introduction

The purpose of this project is to plot the performance three space of a fixed-area missile-class engine with a simple converging nozzle. This will be done by varying the fuel flow rate from 20% to 500% the on design fuel rate, varying the Mach number from 0 to 2.5. Additionally, this will be done at sea level, 4000 m, 8000 m, and 12000 m altitude.

# II. Completely define the on-design engine fluid dynamics and performance

## A. Methodology

## B. Results

# III. Develop the complete performance envelope for this specific turbojet engine

## A. Methodology

## B. Results

*1. Sea Level*

*2. 4000 m*

*3. 8000 m*

*4. 12000 m*

**Table 1 On-Design Operation**

| | | Station | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| $u$ | $m/s$ | 261.87 | 261.87 | 261.87 | 0 | 0 | 0 | 0 | 0 | 0 |
| $P$ | $Pa$ | 35651 | 35651 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| $T$ | $K$ | 236.23 | 236.23 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| $M$ | | 0.85 | 0.85 | 0.50 | 1.0 | | | | | 1.0 |
| $\rho$ | $kg/m^3$ | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| $A$ | $m^2$ | 0.00318 | 0.00318 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| $P_t$ | $Pa$ | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| $T_t$ | $K$ | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

# IV. Conclusion

# Appendix